

Bubble nucleation and dynamics in acoustic droplet vaporization: a review of concepts, applications, and new directions

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Abstract

The development of phase-shift droplets has broadened the scope of ultrasound-based biomedical applications. When subjected to sufficient acoustic pressures, the perfluorocarbon phase in phase-shift droplets undergoes a phase-transition to a gaseous state. This phenomenon, termed acoustic droplet vaporization (ADV), has been the subject of substantial research over the last two decades with great progress made in design of phase-shift droplets, fundamental physics of bubble nucleation and dynamics, and applications. Here, we review experimental approaches, carried out via high-speed microscopy, as well as theoretical models that have been proposed to study the fundamental physics of ADV including vapor nucleation and ADV-induced bubble dynamics. In addition, we highlight new developments of ADV in tissue regeneration, which is a relatively recently exploited application. We conclude this review with future opportunities of ADV for advanced applications such as in situ microrheology and pressure estimation.

Keywords: Acoustic droplet vaporization; Bubble dynamics; Phase-shift droplets; Ultrasound; Microrheology; High-speed microscopy

1 Introduction

Over the past two decades, phase-shift droplets have been investigated in a wide range of biomedical applications. Phase-shift droplets, in their simplest form, are shell-stabilized droplets of a perfluorocarbon liquid. Compared to conventional, ultrasound contrast agents (i.e., microbubbles), phase-shift droplets exhibit enhanced stability, longer lifespan *in vivo*, and the ability to be made in sub-micron sizes. These droplets undergo a phase transition from liquid to gas in response to ultrasound, termed acoustic droplet vaporization (ADV) [1,2], in a non-invasive and spatiotemporally controlled manner. The perfluorocarbon liquid phase in phase-shift droplets enables a thermodynamically favor-

able phase-change due to its high vapor pressure and low surface tension. For example, the vapor pressure of perfluorocarbon liquids is an order of magnitude higher than water [3]. Ultrasound can trigger a phase transition, without substantial generation of heat, when the rarefactional component of the ultrasound wave reduces the local pressure below the saturation vapor pressure of the perfluorocarbon liquid, thereby making vaporization thermodynamically favorable. Since the introduction of ADV, substantial research has been dedicated to the design and fabrication of droplets as well as their interactions with ultrasound at varying acoustic and ambient conditions. Phase-shift droplets of varying formulations have been introduced by using different bulk boiling point perfluorocarbon liquids and shell encapsulations.

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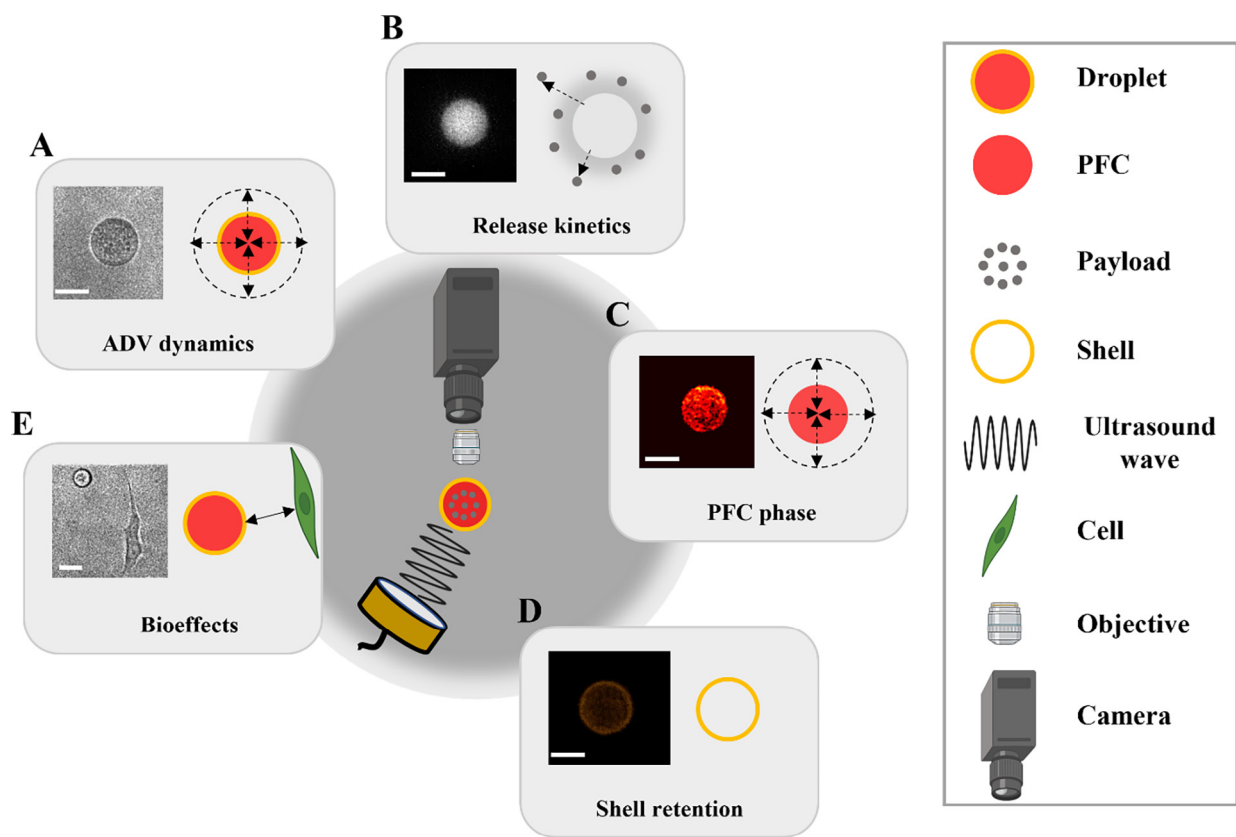


Figure 1. Capturing the dynamics of acoustic droplet vaporization (ADV) of a phase-shift droplet requires ultra-high-speed microscopy. ADV dynamics (A) have been typically studied via brightfield microscopy in a back illumination configuration using a high intensity flash lamp. The study of how ADV impacts payload release in a droplet with a double emulsion morphology (B), the perfluorocarbon (PFC) phase of the droplet (C), the shell of the droplet (D) as well as ADV-generated cellular bioeffects (E) are performed via fluorescence microscopy in a front illumination configuration using a high intensity pulsed or continuous laser. Brightfield and fluorescent images shown in A-E were taken by an ultra-high-speed camera (SIM802 model, Specialised Imaging Ltd. Pitstone, UK). Scale bar: 15 μm . All schematics in this article were created with BioRender.com.

Table 1
Thermophysical properties of commonly used perfluorocarbons for diagnostic and therapeutic applications of ultrasound. Values were taken from Synquest Laboratories, National Institute of Standards and Technology chemistry webbook and Freire, Carvalho [37]. Properties include molecular mass (M), mass density (ρ), surface tension (σ), vapor pressure (p_v), and bulk boiling point (T) at 25°C.

Compound name (Chemical formula)	Acronym	M (kg/mol)	ρ (kg m ⁻³)	σ (mN m ⁻¹)	p_v (kPa)	T (°C)
Perfluoropropane* (C ₃ F ₈)	OFP	0.188	1352	N.A.	793	-39
Perfluorobutane* (C ₄ F ₁₀)	PFB	0.238	1517	N.A.	228	-2
Perfluoropentane (C ₅ F ₁₂)	PFP	0.288	1664	9.5	87	29
Perfluorohexane (C ₆ F ₁₄)	PFH	0.388	1680	12.2	30	56-59
Perfluorooctane (C ₈ F ₁₈)	PFO	0.438	1760	14.4	6	103

* Indicates perfluorocarbon exists in gaseous state at room temperature.

Table 1 contains a list of commonly used perfluorocarbons and their thermophysical properties. Perfluoropentane (PFP) has been frequently used as the perfluorocarbon phase since its bulk boiling point is close to body temperature. However, recent techniques have enabled production of droplets with much lower bulk boiling points that remain meta-

stable in the liquid state under physiological conditions. Several studies have also explored the use of droplets with a bulk boiling point as high as 103°C for tissue regeneration applications (Table 1). To lower the ADV threshold some droplet formulations have been designed to incorporate nucleation seeds such as quantum dots and iron oxide

nanoparticles [4–6]. Droplet formulation and fabrication methods are well developed and extensively reviewed in several previously published articles [7–10].

Understanding the response of a phase-shift droplet to ultrasound is paramount for any ADV-assisted application. Therefore, substantial work has focused on elucidating phase-shift droplet behavior in terms of the acoustic parameters (e.g., frequency and pressure) and droplet thermophysical properties (e.g., size and perfluorocarbon bulk boiling point) [11]. The effects of critical parameters on the ADV threshold has been covered in a prior review article [12]. To characterize the response of phase-shift droplets, both optical and acoustic methods have been employed. Acoustic approaches include recording the acoustic emissions from droplets before and after ADV via active or passive cavitation methods as well as B-mode imaging [13]. Acoustic techniques can detect resonance-sized, bubble oscillations or clouds of bubbles [14,15] but lack the spatiotemporal sensitivity to detect generated vapor nuclei in the early stage of ADV. In addition, conventional methods of producing phase-shift droplets result in polydisperse size distributions. Because vapor nucleation is a stochastic process [16], and the ADV threshold is inversely correlated with droplet size [17], the presence of a small amount of large sized phase-shift droplets can highly skew acoustic measurements performed on populations of droplets. Therefore, direct observations using high-speed imaging are instrumental to assess the effect of various parameters on the complex dynamic of ADV and further elucidate the underlying physics. High-speed, brightfield microscopy has enabled study of ADV at high temporal resolutions, from the inception of the vapor nucleus at nanosecond timescales to the stability and growth of the generated bubbles milliseconds post ADV. In addition to ADV bubble dynamics, recent studies have employed high-speed fluorescence microscopy assisted by high intensity laser illumination to provide sufficient optical contrast and resolution to visualize and study ADV-induced phenomena such as shell behavior, payload release, and potential cellular bioeffects Fig. 1.

Mathematical models for dynamics of spherically symmetric vapor-gas bubbles have been derived from the Rayleigh-Plesset equation, which describes the motion of a bubble in a pressure field [18]. However, the emergence of phase-shift droplets required new considerations [19,20] that are not captured in the existing models. Along with high-speed studies, great progress has been made in the theoretical framework to predict both the phase-transition and the response of the generated bubble under different acoustic and ambient conditions [21]. Developments in the underlying physics of ADV have enabled optimization of droplets for specific applications. Due to the versatility and tunability of phase-shift droplets, a rapidly growing number of applications both in imaging and therapy have been proposed which

are comprehensively covered in several previously published review articles [3,9,22–26].

This review article is composed of four main sections. We review experimental studies utilizing high-speed imaging, and theoretical approaches that have been undertaken over the past two decades to further understand ADV phenomena in Sections 2 and 3, respectively. In Section 4, we highlight how understanding the fundamentals of ADV has benefitted the field of tissue regeneration. Finally, a detailed discussion on potential future applications of ADV, as an *in situ* microrheometer and pressure sensor, is provided in Section 5 of this article.

2 High-speed imaging studies

2.1 Nucleation inception and vapor bubble growth

Although it was initially hypothesized that nucleation originated external to the phase-shift droplet (i.e., heterogeneous nucleation) based on the recorded acoustic cavitation noise [27], direct visualization of nucleation via high-speed microscopy refuted this hypothesis. Several high-speed studies, at a nanosecond timescale, showed that ADV was initiated by homogeneous nucleation within the droplet along the axis of ultrasound propagation [28–30]. The onset of vaporization appeared as either spot-like nucleation sites or homogeneously throughout the droplet possibly due to the limited temporal resolution or the ratio of acoustic focal dimension to droplet size. Earlier high-speed microscopy studies focused on nucleation dynamics during the initial phase-transition within a nanosecond timescale. Kripfgans, Fabiilli [28] showed that ADV began with the formation of a single nucleation site, distal to the ultrasound source, followed by a second nucleus proximal to the source. The authors measured droplet deformation and displacement due to the acoustic wave and showed that ADV occurred once a deformation threshold was reached, independent of the original size. The formation of the initial nucleation site was further investigated by Shpak, Verweij [20] and Li, Kripfgans [29]. Shpak, Verweij [20] mapped the location of the initial nucleation site as a function of the size of the droplet. The nucleation sites were more scattered for droplets smaller than 10 μm in diameter, and highly localized, in the hemisphere proximal to the ultrasound source, for larger phase-shift droplets. Li, Kripfgans [29] extended these studies, demonstrating that the size of the droplet and also the frequency of the incoming wave were critical for vapor nucleation. The authors observed a transition in the initial nucleation site from proximal to distal from the ultrasound source at a critical limit when the droplet size was similar to or larger than the incoming wavelength in PFP liquid (speed of sound: 405 m/s, wavelength at 7.5 MHz: 54 μm). Correlating the location of the initial nucleation site as well as the subse-

quent radial dynamics of the generated bubble with pressure and phase information of the ultrasound pulse provides significant insight into the underlying mechanisms. Shpak, Kokhuis [30] proposed a method of pressure and phase restoration by combining the radial dynamics, obtained at a high sampling rate via high-speed microscopy, and the Rayleigh-Plesset equation. The authors demonstrated that the onset of nucleation occurred during the peak negative half cycle of the ultrasound pulse and shifted towards the end of the pulse.

In addition to nucleation dynamics during the initial phase-transition, high-speed microscopy captured the interaction of the generated vapor bubble with the ultrasound wave as well as its growth dynamics after ultrasound was turned off. Shpak, Kokhuis [30] categorized the dynamics in three distinct regimes: (i) oscillatory translations of the PFP droplet within the surrounding fluid, with amplitudes on the order of 100 nm along the propagation direction of ultrasound prior to nucleation, (ii) rapid growth of the PFP vapor bubble enhanced by rectified heat transfer during ultrasound, and (iii) a square root dependent growth rate up to 3 μ s post ultrasound. Wong, Kripfgans [31] looked at PFP bubble growth during and up to 500 μ s post ultrasound inside a rigid flow tube. Viscous resistance from the tube significantly dampened oscillations of ADV-generated PFP bubbles even though the tube diameter was larger than the bubbles (initial diameters of droplets varied from 0.08 to 0.2 times the tube diameter). Their experimental data suggested four distinct growth stages following nucleation of PFP vapor: (i) a fast linear growth within a microsecond followed by (ii) a slower linear growth over several microseconds and (iii) a non-linear growth up to 200 μ s before reaching (iv) an asymptotic growth. Although high-speed microscopy has provided convincing evidence that ADV is initiated by dynamics independent from acoustic cavitation of the bulk fluid, the residual fluid inertia generated from ultrasound-bubble interactions can impact the response of the ADV-induced bubbles. Li, Kripfgans [32] captured the formation of a bubble torus microseconds after ultrasound was turned off, which was mitigated by reducing the pulse length and driving pressure.

Most high-speed studies have focused on individual (i.e., a single particle in the field of view) micron-size superheated (i.e., above their bulk boiling point) phase-shift droplets, to better understand the ultrasound-droplet interaction (see Table 2). However, since *in vivo* studies rarely rely on activation of a single droplet, studies needed to account for the impact of interparticle interactions on ADV dynamics. Sheeran, Matsunaga [33] studied ADV in concentrated samples of submicron-size perfluorobutane droplets. Higher pressures and longer pulses increased the possibility for secondary effects such as bubble coalescence, destruction, and radiation force interactions in the concentrated samples.

Haworth and Kripfgans [34] reported coalescence of ADV-generated PFP bubbles when a longer pulse was applied, likely because of translational motion due to secondary radiation forces rather than bubbles growing into each other. Coalescence of bubbles may increase the chance of bubble survival for submicron-size droplets by reducing the effect of Laplace pressure and increasing surfactant coating concentration. In another study by Sheeran, Matsunaga [35], the expansion velocity, maximum overexpansion with respect to the initial size of the droplet, and final bubble size were shown to increase with perfluorocarbon volatility (higher degree of superheat). Overexpansion was shown to be dependent on perfluorocarbon volatility (perfluoropropane: $\sim$ 8.3-fold, perfluorobutane: $\sim$ 6.3-fold) and was also more pronounced for larger droplets, since the internal pressure of the gas core due to surface tension was less. Such overexpansion behavior (up to 8 times) was also observed for micron-sized PFP droplets during ADV, followed by damped oscillations [36]. In this study, the ADV-generated bubbles settled to stable diameters much smaller than the maximum size reached during the overexpansion phase. For example, a PFP droplet with an initial diameter of $\sim$ 6 μ m reached a diameter of $\sim$ 40 μ m during a 3-cycle burst and settled to a smaller size bubble ($\sim$ 28 μ m in diameter) after ultrasound was turned off.

As outlined above, particular attention has been given in the literature to the ADV dynamics of low bulk boiling point phase-shift droplets due to the inverse correlation between the ADV threshold and the degree of superheat, which impacts their suitability for diagnostic applications. For a micron-size PFP droplet, the contribution of Laplace pressure will be minimal for the generated bubbles. Furthermore, due to the high vapor pressure (87 kPa at 25°C, 100 kPa at 29°C, and 135 kPa at 37°C), ultrasound triggered stable bubble formation even when studies were conducted at room temperature (i.e., below the bulk boiling point of PFP) [34]. Therefore, achieving different ADV dynamics might be a challenge for micron-size PFP droplets at temperatures and ambient pressures relevant to most biological systems.

However, for drug delivery applications, higher molecular weight perfluorocarbons such as perfluorohexane or perfluorooctane have been utilized due to much enhanced thermal stability [38,39]. Aliabouzar, Kripfgans [40] showed that unique ADV dynamics were achieved using different acoustic conditions and perfluorocarbon droplets (Fig. 2). For drug delivery applications, phase-shift droplets with a double emulsion morphology are typically used, where the deliverable payload is contained within the innermost water phase. Four distinct ADV dynamics for different perfluorocarbon droplets, with a double emulsion structure, embedded within fibrin-based hydrogels were observed: (i) stable bubble formation without recondensation (Fig. 2A), (ii) stable bubble formation with recondensation (Fig. 2B), (iii) tran-

Table 2

High speed brightfield microscopy has been utilized to study dynamics of acoustic droplet vaporization (ADV) for different perfluorocarbon (PFC) phase-shift droplets at various acoustic parameters listed below. Fluorescently labeled shell, PFC phase, payloads, and cells were imaged using high-speed fluorescence microscopy to further investigate ADV-induced phenomena. PFP-perfluoropentane, PFB-perfluorobutane, PFH-perfluorohexane, PFO-perfluorooctane, OFP-perfluoropropane.

High-speed microscopy	Reference	PFC	Size (μm)	Frequency (MHz)	Frame rate (Mfps)
ADV dynamics	Kripfgans, Fabiilli [28]	PFP	5-27	3, 4, 10	5-10
	Haworth and Kripfgans [34]	PFP	3-20	3.5	13
	Wong, Kripfgans [31]	PFP	15-40	3.5	N.A.
	Shpak, Kokhuis [30]	PFP	3.5-11	3.5	18.5
	Sheeran, Matsunaga [33]	PFB	0.19 ± 0.8	1, 5.5, 8	1-20
	Sheeran, Matsunaga [35]	PFB, OFP	2-3	8	20
	Li, Kripfgans [29]	PFP	18	7.5	N.A.
	Li, Kripfgans [32]	PFP	9.1 ± 1.2	7.5	N.A.
	Fan, Kao [64]	PFP	3-6	5	0.5
Shell	Aliabouzar, Kripfgans, et al. [40]	PFP, PFH, PFO	10.1 ± 0.7	1.1, 2.5, 8.6	10-20
	Reznik, Seo [49]	PFP	5-10	10	6×10^{-5}
PFC phase	Reznik, Seo [49]	PFP	5-10	10	6×10^{-5}
	Aliabouzar, Kripfgans, et al. [40]	PFP, PFH	15.1 ± 1	2.5	0.2-0.5
Payload	Aliabouzar, Kripfgans, et al. [40]	PFP, PFH, PFO	15.1 ± 1	2.5	0.5-1
	Wang, Kang [65]	PFP	0.48 ± 0.04	3.5	0.1
Bioeffects	Kang, Lin [60]	PFP	2.1 ± 0.2	2	3.6×10^{-3}
	Qin, Zhang [61]	PFP	0.6 ± 0.03	5	0.2
	Fan, Lin [62]	PFP	1.5-3	5	0.2-0.5
	Feng, Qin [63]	PFP	1.1 ± 0.3	5	0.2

sient bubble formation with recondensation (Fig. 2C), and (iv) fragmentation (Fig. 2D). The authors recorded ADV-induced stable bubble formation in micron-sized superheated PFP- and subcooled perfluorohexane phase-shift droplets, although the growth rate of the latter was an order of magnitude lower after ultrasound. Due to the significantly higher supersaturation ratio (i.e., pressure inside the ADV-generated bubble to the equilibrium vapor pressure), perfluorooctane phase-shift droplets at any acoustic parameters did not form stable bubbles.

Data obtained from high-speed imaging, as summarized above, has provided detailed visualization of ADV at high temporal resolution and enhanced fundamental understanding of the underlying mechanisms. However, there is a need to develop more advanced experimental techniques to understand ADV mechanisms in an *in vivo* setting. Furthermore, high-speed imaging utilizes widefield microscopy which is limited to the focal plane (i.e., 2D) and is prone to out-of-plane distortion and blurring. Combining it with confocal microscopy may help reduce these limiting factors [41]. In the following section we review the studies conducted with high-speed fluorescence microscopy.

2.2 Shell stability

Phase-shift droplets stabilized with albumin or phospholipids have been typically used in non-drug delivery applications of ADV [4,13], while polymer (e.g., Poloxamer)

stabilized phase-shift droplets have been primarily used in drug delivery applications [24,25,42]. The shell plays a key role in maintaining the stability of the droplets by lowering the interfacial tension and gas permeability, as well as impacting the linear and nonlinear acoustic responses of the generated bubbles. Shell behavior under ultrasound exposure (e.g., shedding, rupture, and buckling) is a well-studied phenomenon with contrast microbubbles [43,44] and can lead to different bubble response characteristics including thresholding behavior, compression-only behavior, and subharmonics [18,45-47]. High-speed fluorescence imaging at frame rates up to 150,000 frames per second has captured dynamics of lipid shedding in contrast microbubbles at varying acoustic pressures and pulse durations [48]. For phase-shift droplets, shell properties have not been fully explored during and post ADV, particularly in cases with a reversible phase-transition (i.e., repeated vaporization and recondensation).

Reznik, Seo [49] studied shell retention and growth rate of PFP droplets with three different fluorescently labeled shells. For a lipid-coated PFP droplet, the fluorescence signal remained homogeneously distributed on the surface of the newly generated PFP bubble for up to 7 s post ADV, whereas heterogeneous patterns were observed for albumin and polymer coated phase-shift droplets. The authors reported no rapid dissolution for the newly generated bubbles with different shells for up to 15 min. However, the growth rate of PFP bubbles with a lipid shell was up to five

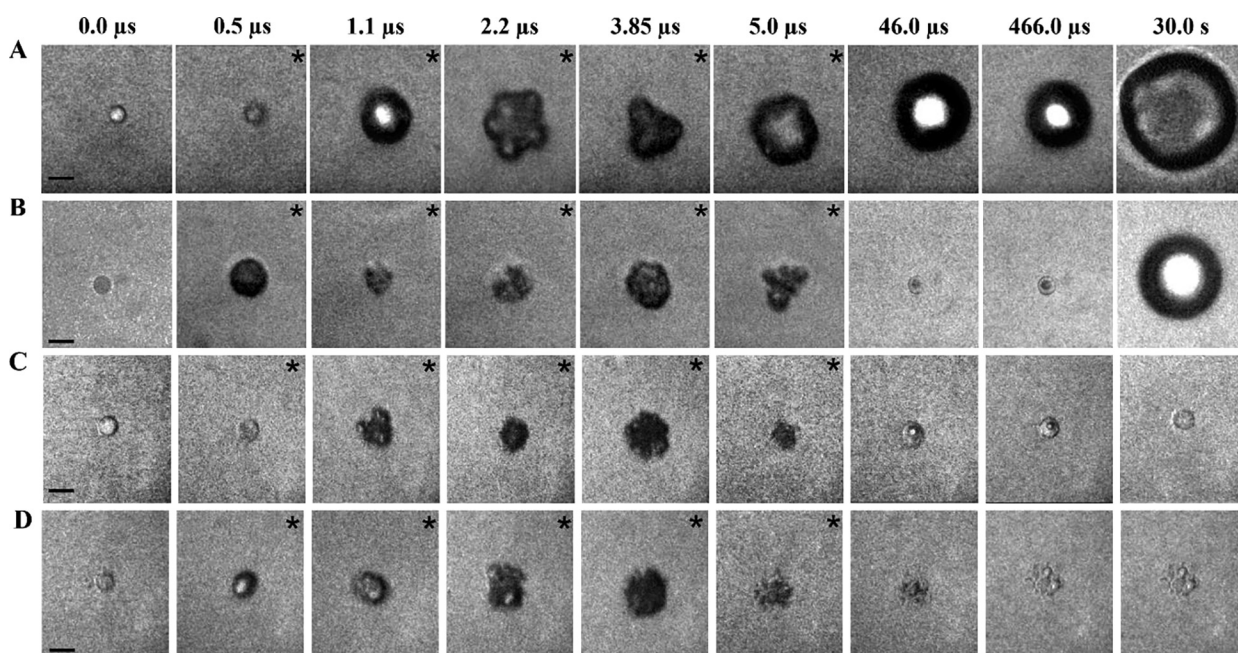


Figure 2. Ultra-high-speed microscopy captured the dynamics of acoustic droplet vaporization for different phase-shift droplets before, during, and after ultrasound exposure: perfluoropentane (PFP), perfluorohexane (PFH), and perfluorooctane (PFO). Droplets were embedded in 10 mg/mL fibrin hydrogels. Four distinct ADV dynamics were observed: stable bubble formation without recondensation (PFP) (A), stable bubble formation with recondensation (PFH) (B), transient bubble formation with recondensation (PFO) (C), and fragmentation (PFD) (D). Acoustic parameters were as follows: a single burst of 6 μ s, 2.5 MHz, and 6.5 MPa peak rarefactional pressure. An exposure time of 100 ns was used. Frames containing ultrasound are denoted with an asterisk. Scale bar: 15 μ m. Panels A-D reprinted from Aliabouzar, Kripfgans [40], with permission from Elsevier.

times slower due to shell retention and thus shell-induced resistance to gas transfer. The state of the shell at different acoustic settings during ADV and much longer time points post-ADV is currently unknown [50]. Future studies are needed to optimize shell performance, either retention for diagnostic applications or fast dissolution which may be beneficial for therapeutic applications of ADV such as drug delivery.

2.3 Perfluorocarbon phase dynamics

Fluorescent markers have been used to selectively label the perfluorocarbon phase of the phase-shift droplet. Homogeneous distribution of fluorescently labeled PFP, which was localized within the droplet, was disrupted following ADV [40,49]. For a PFP droplet, the fluorescently labeled dye remained at the interface between the ADV-generated bubble and the surrounding medium, likely due to its hydrophobic nature. However, for a perfluorohexane droplet, high speed fluorescence microscopy revealed recondensation of the perfluorohexane liquid microseconds after ultrasound was turned off [40].

2.4 Payload release kinetics

Phase-shift droplets present a promising alternative to other ultrasound-based drug delivery platforms such as microbubbles both in terms of stability and greater drug-loading capacity, resulting in therapeutically relevant dosages. Drugs can be loaded into the liquid core of the phase-shift droplets compared with microbubbles where drugs are loaded into the shell [51]. Microfluidic techniques have been used to generate monodisperse, payload-carrying phase-shift droplets with a double emulsion morphology where the payload was within the innermost phase [52,53]. With a double emulsion morphology, an encapsulation efficiency of 99% can be achieved [52] while retaining the ability to trigger at similar acoustic pressures. Although several ADV-assisted drug delivery studies have been conducted, both *in vitro* and *in vivo* [52–56], the dynamics of release have not been visualized and characterized before. One challenge is to select bright and efficient fluorophores to permit optical visualization at high frame rates relevant to ADV. Pulsed lasers are used to excite the fluorophores, however illumination repetition rate is limited. Aliabouzar, Kripfgans

[40] characterized the release dynamics at multiple time-scales (i.e., pre-, during, and post-ADV). This study showed that significantly different release kinetics were achieved when phase-shift droplets of varying bulk boiling points were used. ADV-induced stable bubble formation, which occurred for lower bulk boiling point phase-shift droplets, was associated with complete release of the fluorescently labeled payload, in a form of cloud release. Transient bubble formation, which occurred for higher bulk boiling point phase-shift droplets, resulted in partial release. ADV-induced stable and transient bubble formation has provided a unique platform to modulate release dose and kinetics in a spatiotemporally controlled manner, as will be discussed later in [Section 4](#).

2.5 Cellular and mechanical bioeffects

Determination of spatiotemporal bioeffects of ADV is critical for developing safe and effective diagnostic and therapeutic applications. The occurrence of ADV is accompanied by mechanical forces acting on the surrounding tissues through the rapid volumetric expansion during the phase-transition on microsecond timescales, and subsequent oscillations of the generated bubble during the remaining ultrasound pulses. High bubble interface velocities on the order of tens of m/s [30], and the associated normal (up to 90 kPa) and shear (up to 8 Pa) stresses [57] may give rise to a range of desired or undesired bioeffects. Furthermore, the authors showed that ADV-induced stresses significantly depended on the initial size of the droplet and viscosity of the surrounding fluid. The calculated shear stresses fall within the range reported in human arteries during basal conditions (0.2–2 Pa rising to 10 Pa near arterial branches) [58].

High-speed microscopy has been used to visualize and quantify ADV-associated mechanical and cellular bioeffects on fluorescently labeled cells and vessel phantoms. Wang, Kang [59] showed that conjugated PFP droplets induced mechanical forces to disrupt T lymphoblast membranes, upon exposure to an ADV pulse, evident by cell debris and spilled cytoplasm and organelles. The integrity of the fluorescently labeled cell membrane did not appear to change significantly with ADV of a single droplet of PFP, indicating that the severity of cell membrane damage was dependent on the number of targeted phase-shift droplets attached to the cell. Kang, Lin [60] recorded the morphological characteristics and progression of ADV-induced lesions in an agarose-based, vessel phantom, with a diameter much larger than the generated PFP bubble, at varying phase-shift droplet concentrations, phantom stiffnesses, and acoustic parameters. The occurrence of phantom erosion and tunnel-shaped lesions increased with the ultrasound driving pressure, phase-shift droplet concentration, and number of cycles. At low droplet concentrations, shallow erosions were

observed at both sides of the vessel wall. However, at higher phase-shift droplet concentrations, much deeper erosions in the form of tunneling occurred on the side of the wall proximal to the transducer pointing to acoustic shielding of the generated bubbles at higher concentrations. Using a passive cavitation detection technique, the authors correlated the initiation of lesion formation with inertial cavitation.

Spatial mapping of ADV-induced bioeffects has provided further insight into bubble-cell interactions and real-time cell response [61,62]. Membrane permeability and viability of C6 glioma cells as a function of droplet-to-cell distance, acoustic pressure, and droplet size was characterized. The observed cellular bioeffects depended on the droplet-to-cell distance (d) and were categorized into three types for a droplet of $\sim 3 \mu\text{m}$ in diameter and peak rarefactional pressure of 5 MPa: (i) intact membrane and normal viability (d : $16.9 \pm 7.4 \mu\text{m}$), (ii) reversible membrane permeability enhancement and normal viability (d : $9.9 \pm 1.9 \mu\text{m}$), and (iii) irreversible membrane permeability and loss of viability (d : $3.8 \pm 1.7 \mu\text{m}$). The measured threshold distances decreased significantly as the excitation pressure increased for all three categories. Spatial mapping of ADV-induced, cellular bioeffects indicated that the maximum distance at which the bioeffects were observed correlated with the maximum radial expansion of the generated bubbles during ADV. Furthermore, the most severe bioeffect (i.e., type iii) occurred when the droplet-to-cell distance was smaller than the resting size of the generated bubble. Although this study focused on a PFP droplet, the findings suggest that higher bulk boiling point perfluorocarbons may cause shorter range bioeffects since the maximum radial expansion during ADV was shown to correlate inversely with the bulk boiling point of droplet [40]. High-speed microscopy was also used to monitor ADV-bubble growth, coalescence, lodging, and subsequent capillary occlusion or rupture in an *ex vivo* rat model [63]. ADV induced transient microvessel wall invagination locally, which returned to the original state following a relaxation process. Three characteristic behaviors were recorded for ADV bubbles in the microvessels: (i) translational movement, (ii) aggregation, and (iii) coalescence. [Table 2](#) summarizes experimental studies in [Section 2](#) using high-speed microscopy to investigate ADV dynamics and ADV-induced phenomena.

3 Models developed for ADV

3.1 Prediction of phase-shift droplet expansion

Perfluorocarbon droplets will have different thermophysical properties, in comparison to the bulk phase. Phase-shift droplet size, in particular, was shown theoretically to impact solubility, effective boiling point, and the size change associated with vaporization [66]. The ADV-induced expansion

factor, defined as the ratio of an expanded bubble during ADV to the corresponding initial droplet diameter, varied greatly depending on the initial diameter of the phase-shift droplet. Using the ideal gas law and the Young-Laplace equation, along with certain assumptions such as negligible gas diffusion during phase conversion, the expansion factor for varying phase-shift droplet radii was derived [17,66]. An expansion factor of approximately 5 was reported for a sufficiently sized droplet [28,67,68] in water. As such, it was shown that the confinement effect of Laplace overpressure was significant for droplets below 3 μm in radius, resulting in an expansion factor smaller than 5. In addition to the initial size of the droplet, retention of the shell post ADV can significantly impact the interfacial surface tension and thereby the Laplace pressure. For the same initial droplet size, the expansion factor was higher assuming shell retention (i.e., interfacial surface tension: 25 mN/m) compared to a free bubble (i.e., interfacial surface tension: 72 mN/m) [69]. Pitt et al. investigated nanodroplets of perfluorocarbon, specifically physical aspects of gas exchange and dependence on acoustic frequency, sound pressure, heat etc. [72].

3.2 Prediction of ADV-induced normal and shear stresses

As discussed above, several factors influence ADV phenomena. Therefore, these factors need to be included in models predicting the phase-transition and the response of the generated bubble. Earlier models of ADV were motivated by the application of gas embolotherapy (i.e., the occlusion of a blood vessel using an ADV-generated bubble), thus aiming to estimate the mechanical stresses induced by the expanding bubbles on the walls of blood vessels. Velocity and stress distributions of the expanding bubbles, without modeling the phase-change, were analyzed in rigid [70] and flexible [71] tubes for varying bubble to tube diameter ratios of 0.1–0.9. The calculated normal pressure was highest at early stages of expansion and decayed within a few microseconds. However, the induced wall shear stress peaked at a later stage of expansion, as the bubble approached the wall, and lasted longer. The ADV-induced wall stresses correlated directly with the size of the bubbles and were higher in a rigid tube compared to a flexible tube. In another study from the same group, stress and velocity distributions were further examined by including the phase-change process [57]. ADV-induced, normal and shear stresses correlated directly with the initial PFP droplet size (7–14 μm) and peaked within the range of 70–90 kPa and 2–8 Pa, respectively.

3.3 Prediction of phase-transition, growth, and stability of ADV bubbles

Several mathematical models have been proposed to simulate the phase-transition and growth dynamics of phase-

shift droplets. PFP bubble evolution during and post ADV was analytically solved using a generalized Rayleigh-Plesset equation, assuming a constant evaporation rate at the liquid-gas interface [68]. The results obtained from their proposed ADV model were in good agreement with the linear growth of the post-ADV PFP bubbles as observed experimentally via high-speed imaging by Wong, Kripfgans [31]. Their proposed model also suggested an oscillatory nature of bubble evolution for PFP droplets smaller than 6 μm in diameter and a smooth, asymptotic evolution for larger droplets. An improved theoretical model predicting growth dynamics of a PFP-bubble including descriptions of thermal diffusion as well as gas diffusion through the bubble surface was initially proposed by assuming a thermal boundary layer theory [21], and later by numerically solving the Keller-Miksis equation accounting for large amplitude oscillations [30]. To consider the influence of ultrasound, their model also included the effect of rectified heat transfer. Their results underlined the fundamental role of gas diffusion during ADV, suggesting that a minimum of 10% of air solubility in PFP was necessary to allow the bubble to survive the first collapse in the compressional wave. Outside the context of ADV, several models have been proposed to indicate the importance of gas diffusion for microbubble stability [73,74].

In additional studies, Shpak, Verweij [20] and Li, Kripfgans [29] provided experimental and theoretical results suggesting that ADV was initiated by the focusing of a nonlinear acoustic wave on a specific spot inside the phase-shift droplet. The focusing resulted from the spherical shape of the droplet, low speed of sound in perfluorocarbon liquids (i.e., significant change in acoustic impedance relative to the surrounding water medium), and the presence of higher harmonics due to nonlinear propagation of the wave. The superharmonic focusing effect generated a significant negative pressure inside the phase-shift droplet (~ 6 -fold increase in peak negative pressure) which correlated with the position of initial vapor nucleation in the droplet [75]. The amount of focusing is a close interplay between the degree of constructive interference of the higher harmonics within the droplet and the size of the droplet. In addition to acoustic parameters such as frequency and pressure, the shape of the nonlinearly distorted wave is strongly dependent on the properties of the propagating medium [76]. A measure of the likelihood of nonlinear distortion is the Goldberg number, defined as the ratio of nonlinear distortion to attenuation [77]. The Goldberg ratio is much higher for water, which is by comparison a significantly less attenuating and much more nonlinear medium than tissues. For example, at 5 MHz and 5 MPa, the Goldberg ratio is 266 and 12 for water and fibrin gel (i.e., similar to soft tissues such as liver and breast), respectively [78]. Thus, nonlinear distortion is harder to achieve in tissue than in water for sim-

ilar acoustic parameters. Additionally, phase-shift droplets have high surface reflectivity depending on the choice of perfluorocarbon. Aliabouzar, Lu [78] showed that the amplitude of higher harmonics at higher frequencies was diminished due to increased overall scattering in fibrin gels containing higher volume fractions of perfluorohexane droplets, thereby potentially inhibiting superharmonic focusing.

3.4 Effects of degree of superheat and encapsulation on the ADV process

The models described above assumed the evolution of ultrasound-driven perfluorocarbon vapor inside a superheated infinite medium of the same perfluorocarbon liquid. This assumption limited the study of conditions such as complete conversion of a droplet into a bubble or overexpansion. Doinikov, Sheeran [79] proposed a theoretical model describing the growth of a vapor bubble inside a finite size droplet until complete conversion, and subsequent overexpansion and damped oscillation of the generated vapor bubble until settling to a final diameter. Their model predicted that the degree of superheat correlated inversely with the time necessary for the same size droplet to fully vaporize (perfluoropropane: $\sim 0.5 \mu\text{s}$, perfluorobutane: $\sim 1 \mu\text{s}$, and PFP: $\sim 6 \mu\text{s}$) and directly with the overexpansion of the vapor bubble (perfluoropropane: ~ 7 , and perfluorobutane: ~ 6). Although the difference between boiling points for perfluorobutane and perfluoropropane is similar to that for perfluorobutane and PFP (see Table 1), the difference in the calculated time of complete vaporization was significantly different, indicating a nonlinear vaporization dynamic for droplets. Their model was further validated by high-speed microscopy of perfluoropropane and perfluorobutane droplets [35]. Expanding further on the finite size of the droplet, Guédra and Coulouvrat [80] proposed a model accounting for shell encapsulation by including the effects of surface tension and shell rigidity using a former model developed for ultrasound contrast agents [81]. Their model showed that the shell slowed the growth of the vapor phase in the first stage of expansion, while a sufficiently rigid shell (e.g., shear elasticity modulus $>400 \text{ MPa}$) prevented vaporization of the PFP droplet. At higher frequencies, the shell resulted in condensation of the vapor phase during the first acoustic period. Higher shell rigidity yielded a shorter condensation phase and a faster collapse of the generated bubble.

3.5 Classical nucleation theory predictions of vaporization and condensation thresholds

In the aforementioned models, the acoustic excitation pressure was given by a simplified harmonic source including the contribution of the static pressure. However, a more realistic model would include effects of nonlinear distortion

and higher harmonic generation due to nonlinear propagation of the wave before reaching the droplet. Miles, Doering [82] combined classical nucleation theory (CNT) with the effects of nonlinear distortion of the wave to predict the ADV threshold of a PFP droplet as a function of parameters such as frequency, droplet size, and ambient temperature. Although CNT is the most used theory of nucleation, it predicts the ADV threshold to be weakly dependent on frequency (i.e., reciprocal of the duration of time that the liquid is under negative pressure), which is inconsistent with experimental results. Lajoinie et al. [83] derived a theoretical description with matching experimental investigations to demonstrate an existing resonance with $kR \sim 0.65$ (i.e., wavenumber (k) times droplet radius (R)), for which a 3-fold pressure amplification inside the droplet causes a significant increase of the nucleation rate. In the context of cavitation in liquids, Fisher [84] showed theoretically that the nucleation pressure only decreases by $\sim 36\%$ if the time that a liquid is under tension increases from 10^{-15} s (i.e., less than the time required for a sound to travel one atomic distance) to 10^{18} s (i.e., the estimated age of the universe). In addition, CNT predicts higher nucleation thresholds compared to experimentally measured ADV thresholds. This discrepancy can be due to the limitation of experimental methods for detecting ADV thresholds, particularly via acoustic techniques, which are relatively insensitive to nano-size vapor nuclei. Longer pulse durations and lower frequencies facilitate growth of nucleated bubbles, making them easier to be detected experimentally. Including the effect of dissolved gas content in the droplet, by accounting for the compositional dependence of surface tension of the perfluorocarbon liquid and the pressure-dependent solubility of the gases in the liquid, significantly reduced the predicted nucleation threshold to levels close to the experimentally measured values [85]. Furthermore, a constant nucleation rate is typically assumed in CNT whereas in the context of cavitation and ADV, the nucleation rate varies during the rarefactional half cycle of the ultrasound wave (i.e., low nucleation rate at the beginning of the rarefactional half cycle and highest nucleation rate at the peak of the rarefactional pressure). Integrating the nucleation rate over the negative half cycle of the ultrasound pulse or using a non-classical nucleation approach could yield better models that are more consistent with experimental data. In a recent study, CNT was modified to include Laplace pressure effects on the thermodynamics of PFP nanodroplets such as effective surface tension of the vapor nucleus, resulting in a larger nucleation rate and ADV thresholds closer to the experimentally measured values [86]. Their modified CNT predicted an ADV threshold of 4.5 MPa for a PFP nanodroplet while CNT, without Laplace pressure consideration, predicted a much higher ADV threshold (i.e., 8.9 MPa), highlighting the inherent limitation of CNT due to the capillary approximation

assumption. In addition to the nucleation threshold, CNT has also been utilized to predict the condensation pressure for the ADV-generated bubbles in droplets with different perfluorocarbon cores, suggesting a direct correlation between the volatility of the perfluorocarbon droplet and the condensation pressure [85]. The authors also reported a direct correlation between the threshold bubble radii above which the ADV-generated bubbles remained stable and the perfluorocarbon bulk boiling point. However, at a significantly high supersaturation ratio, ADV-generated bubbles condensed to liquid following ultrasound exposure irrespective of bubble size.

3.6 Effects of critical parameters on the fate of ADV bubbles

More recent numerical models were developed to investigate the effect of acoustic amplitude, frequency, the number of cycles, and liquid compressibility on the survival, growth, and collapse of ADV-generated bubbles in phase-shift droplets [87–89]. Lacour, Valier-Brasier [88] developed a phase diagram mapping different regimes of ADV as a function of frequency and amplitude. Irreversible collapse and complete vaporization occurred at the lowest and highest driving pressures, respectively. Between the irreversible collapse and complete vaporization, the intermediate regime was shown to be more complex. Depending on the interplay between surface tension, heat transfer, driving pressure and frequency, multiple fates including rebounds took place before complete vaporization. Their models also showed that the intermediate regime was more complex at higher frequencies. Numerical models have accounted for liquid compressibility, which directly impacted the ADV threshold, as well as the time required for complete vaporization of a PFP phase-shift droplet particularly at higher frequencies [90,91]. Another important factor impacting bubble growth following vaporization is dissolved gas content in the surrounding medium. The amount of gas uptake (e.g., N_2 , O_2 , CO_2 , and Ar) from the host liquid into the generated bubble has been modeled [2]. Based on the developed model, a generated PFP bubble of $\sim 17\ \mu m$ in diameter expanded to $100\ \mu m$ in diameter in less than 5 s. In a recent study, Radhakrishnan, Holland [92] developed a numerical model predicting ADV-induced changes in dissolved oxygen content in the surrounding fluid at equilibrium. Reduction in dissolved oxygen content was shown to depend on PFP droplet concentration, both numerically and experimentally, and dropped to as low as 25% of air saturation. An overview of the models developed to study ADV dynamics as well as stability and growth of the generated bubbles is summarized in Fig. 3.

The fundamental findings, both from high-speed imaging and modeling, have enabled optimization of droplets for

specific applications ranging from imaging to therapy. In the following section we discuss how ADV-induced bubble dynamics have been utilized in the context of tissue regeneration.

4 Tuning ADV dynamics in tissue regeneration

4.1 On-demand modulation of critical release parameters

Tissue regeneration is highly dependent on the spatiotemporal presentation of bioactive molecules (e.g., growth factors). Factors such as the concentration, release kinetics, and sequence of growth factors can significantly impact regenerative processes. For example, low to medium concentrations of vascular endothelial growth factor induced growth of stable capillaries with uniform size while above a critical concentration, leaky and abnormal capillaries were formed [93]. Additionally, kinetics of release of growth factors plays a key role in regulating cellular responses. For example, sustained release of basic fibroblast growth factor was more effective than a bolus release in increasing vascular endothelial and smooth muscle cell proliferation [94]. To inhibit vascular cells, a bolus release of transforming growth factor- $\beta 1$ was more effective than the same amount released in a sustained manner.

A commonly used strategy in tissue regeneration is to incorporate exogenous, bioactive payloads in a biocompatible, hydrogel matrix that is implanted at the intended site of regeneration [95,96]. Release of the payload from the matrix is dependent on payload-matrix affinity and matrix degradation [97]. However, this strategy affords limited spatiotemporal control of payload release. ADV is a potential paradigm shift within tissue regeneration due to its ability to spatiotemporally control payload release. For tissue regeneration, phase-shift droplets are typically embedded in a hydrogel matrix to yield a smart platform, termed an acoustically responsive scaffold. For acoustically responsive scaffolds to remain stable after implantation over the course of days or weeks, higher bulk boiling point perfluorocarbons have been used to minimize spontaneous vaporization (i.e., vaporization in the absence of ultrasound) [98]. Spatiotemporal control of critical release parameters can be tuned with ADV dynamics by varying droplet bulk boiling point and acoustic conditions. Different ADV-induced bubble dynamics and the corresponding payload release are schematically shown in Fig. 4A for phase-shift droplets with a double emulsion structure. Along with high-speed microscopy, confocal fluorescence microscopy was used to image and characterize payload-carrying phase-shift droplets before and after ADV, though at a much slower timescale (i.e., hours) [85] (Fig. 4B).

ADV-induced stable (Fig. 4B-II) and transient bubble (Fig. 4B-III) formation correlated with complete and partial

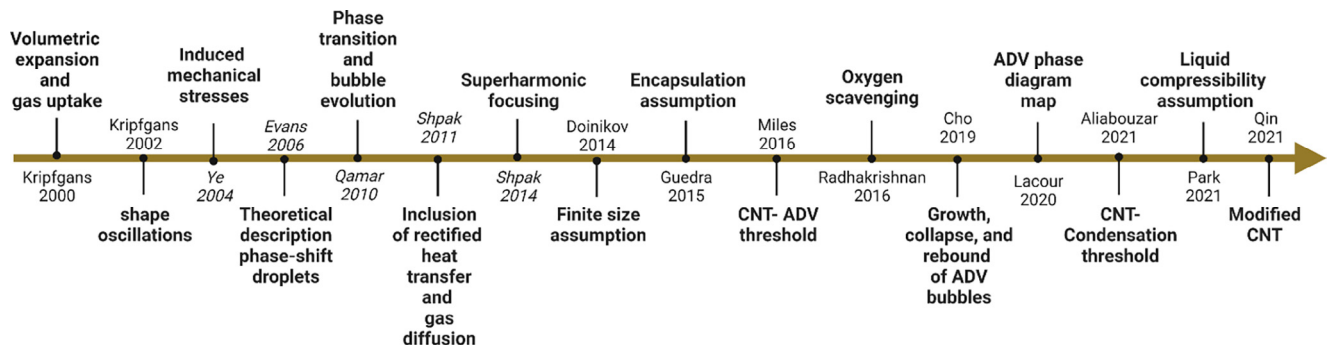


Figure 3. Timeline displaying models developed to study acoustic droplet vaporization (ADV) of a phase-shift droplet as well as stability and growth dynamics of ADV-induced bubbles. Models are indicated by publication date and first author.

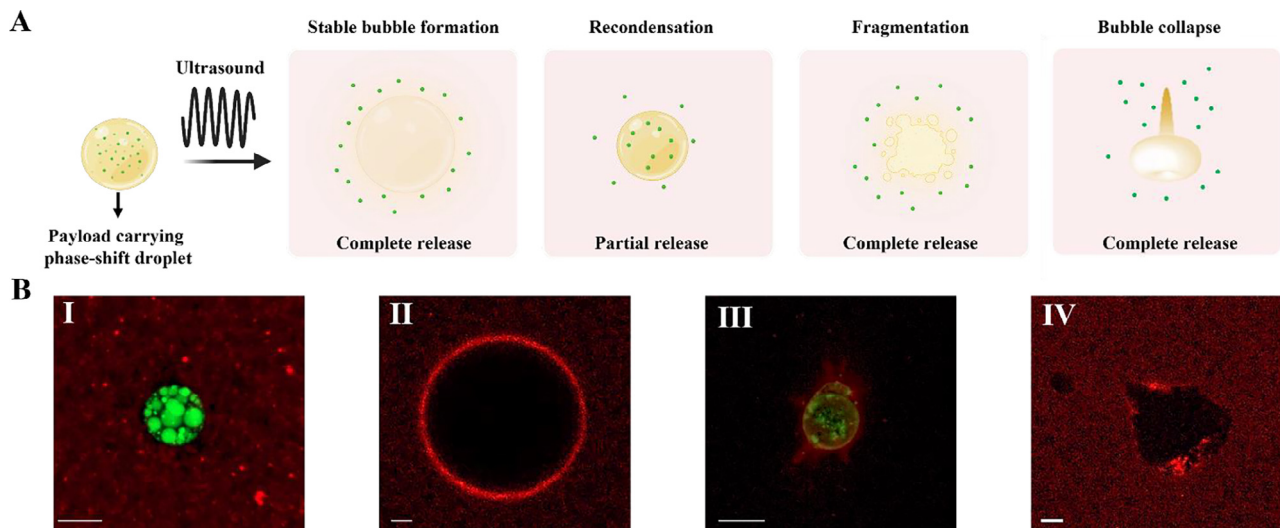


Figure 4. Acoustic droplet vaporization (ADV) enables dynamic modulation of release kinetics from phase-shift droplets with a double emulsion structure. Tuning the acoustic conditions (e.g., frequency) and thermophysical properties of droplets (e.g., bulk boiling point) can result in different bubble dynamics and resulting release kinetics (A). Droplets, loaded with Alexa Flour 488-labeled dextran (green), were embedded in a 5 mg/mL fibrin hydrogel that was labeled with Alexa Flour 647-labeled fibrinogen (red). Confocal microscopy images show that the droplet contained the dextran payload before the arrival of ultrasound (B, I). ADV-induced stable bubble formation completely released the payload for lower bulk boiling point droplets and lower excitation frequencies (B, II). For higher bulk boiling point droplets, repeated vaporization and recondensation partially released the payload (B, III). Other ADV dynamics such as asymmetrical collapse of the bubble, by positioning droplets near a rigid boundary, and fragmentation have been reported which result in complete release of the payload without stable bubble formation (B, IV). Scale bar: 5 μ m. Panel (B, III) reprinted from Aliabouzar, Kripfgans [85], with permission from Elsevier.

payload release, respectively, as confirmed with both high-speed fluorescence imaging and confocal microscopy. ADV-induced collapse and fragmentation have also been associated with complete release of the payload without formation of stable bubbles (Fig. 4B-IV). Stable bubble formation in acoustically responsive scaffolds was used to optimize release kinetics, another critical delivery parameter [99]. For example, release rate was significantly lower in acoustically responsive scaffolds containing perfluorohexane droplets compared to perfluorooctane droplets due to the

presence of ADV-generated bubbles. A longer diffusion path is expected in acoustically responsive scaffolds with stable bubble formation (i.e., in the case of perfluorohexane) since released payloads cannot diffuse through a bubble, but rather must diffuse around it. More advanced techniques such as 3D bioprinting can be utilized to micropattern payload-carrying droplets as well as control the response of the generated bubbles in acoustically-responsive scaffolds [100]. The unique stable or transient bubble formation induced by ADV can be utilized in applications beyond tissue regen-

eration. For example, transient bubble formation can be leveraged for an on-and-off blinking contrast signal between consecutive ultrasound pulses, offering better localization of bubble signal from tissue [101–105].

4.2 On-demand modulation of micromechanics

Several high-speed imaging and theoretical models have extensively characterized the radial expansion of phase-shift droplets with varying bulk boiling points and sizes as outlined in Sections 2 and 3. The volumetric expansion of the perfluorocarbon phase during ADV, successive oscillations post-ADV, and further growth of the generated bubbles due to in-gassing can have significant impact on the surrounding tissue. While the effects of bulk medium properties (e.g., bulk elastic and viscous moduli) on ADV phenomena has been investigated [105–108], the impact of ADV on the surrounding medium has only been recently studied. Modification of dissolved gas content (e.g., nitrogen and oxygen) in surrounding fluids via ADV has been noted by several groups [108,109]. A significant reduction in the partial pressure of oxygen in a fluid, from 100% to 25% relative to air saturation, was achieved by ADV [110], introducing potential new applications in decompression sickness and myocardial infarction. In a recent study, ADV was used as a technique to alter the porosity and stiffness of a fibrin-based acoustically responsive scaffold in a dynamic, spatiotemporally-controlled manner [111]. For a fibrin-based acoustically responsive scaffold, the ultrasound forcing period (i.e., inverse of frequency) is much shorter than the fibrin relaxation time leading to strain build up in the matrix. Since fibrin, like other natural biomaterials, exhibits strain stiffening behavior, ADV-generated bubbles can induce significant micromechanical and microstructural changes. Both bulk and microrheology techniques were utilized to characterize acoustically responsive scaffolds post ADV. Using dynamic shear testing, Fabiilli, Wilson [112] reported a 300% increase in the shear modulus in an acoustically responsive scaffold with stable bubble formation. ADV-induced radial compaction of fibrin at the bubble-fibrin interface was visualized using confocal microscopy [111]. The width and density of the compacted fibrin region increased over time due to diffusion-driven growth of the bubble (Fig. 5). In the same study, ADV-induced stiffening was mapped spatially and longitudinally within acoustically responsive scaffolds via atomic force microscopy. Although the field of ultrasound-induced modulation of physical and micromechanical properties in soft biomaterials is in its infancy, ADV-induced modulation of micromechanical properties highlights the potential of this approach.

On-demand micromechanical changes induced via ADV can have broad biomedical applications and lead to novel approaches. Two recent studies used this approach for on-

demand modulation of cellular signaling in a cancer model [114] (Fig. 6A) and cell differentiation [113] (Fig. 6B).

5 Future Outlook

5.1 ADV as an *in situ* microrheometer

For several decades, quantitative characterization of tissue properties using non-invasive ultrasound has been an exciting research prospect. Elastic and viscous properties of tissues can be characterized by measuring the speed of sound and its attenuation, respectively. However, variation in compressional sound velocity and therefore bulk modulus is significantly less than an order of magnitude among various soft tissues [115]. The limited dynamic range of bulk moduli across different soft tissue types is problematic for tissue characterization via ultrasound-based techniques. Unlike bulk modulus, the shear modulus varies more significantly across different soft tissues. For example, compressional sound velocity in different types of breast tissues varied from 1450 m/s to 1584 m/s [116], while shear wave velocity varied from 1.1 m/s to 9.6 m/s [117]. Recent ultrasound-based elastography techniques rely on measuring shear wave velocity to compute tissue elasticity via a simple conversion of shear modulus to Young's modulus assuming incompressibility of soft tissues [118,119]. However, there are several limitations associated with shear wave elastography techniques including high operator skill dependence, reproducibility, tissue size requirements, and high attenuation of shear waves over a short propagation distance within diagnostic frequencies [120–122].

Recent cavitation-based microrheology techniques, pioneered by Kundu and Crosby [123], have emerged to determine the length- and time-dependent characteristics of soft biomaterials. Since bubbles are extremely sensitive to the surrounding media, changes in bubble responses due to rheological properties of the medium can be significant. **The presence of a soft, viscoelastic medium strongly affects bubble dynamics, both linear resonance properties and nonlinear behavior.** For a detailed review of the effect of medium rheology on bubble dynamics, readers are referred to Dollet, Marmottant [124]. Two bubble-based, mechanical characterization approaches have been proposed in the literature: (i) bubble translation due to applied radiation force and (ii) resonant bubble oscillations. Both direct imaging and measuring acoustic scattering spectrum have been used to record the bubble responses. In the bubble displacement technique, a short pulse of ultrasound pushes the bubble through a medium via acoustic radiation force and its position is tracked over time, thereby revealing rheological properties of the medium [125]. In the oscillating bubble approach, a low amplitude ultrasound wave drives bubble oscillation. Elasticity of the medium is derived by recording a shift in the res-

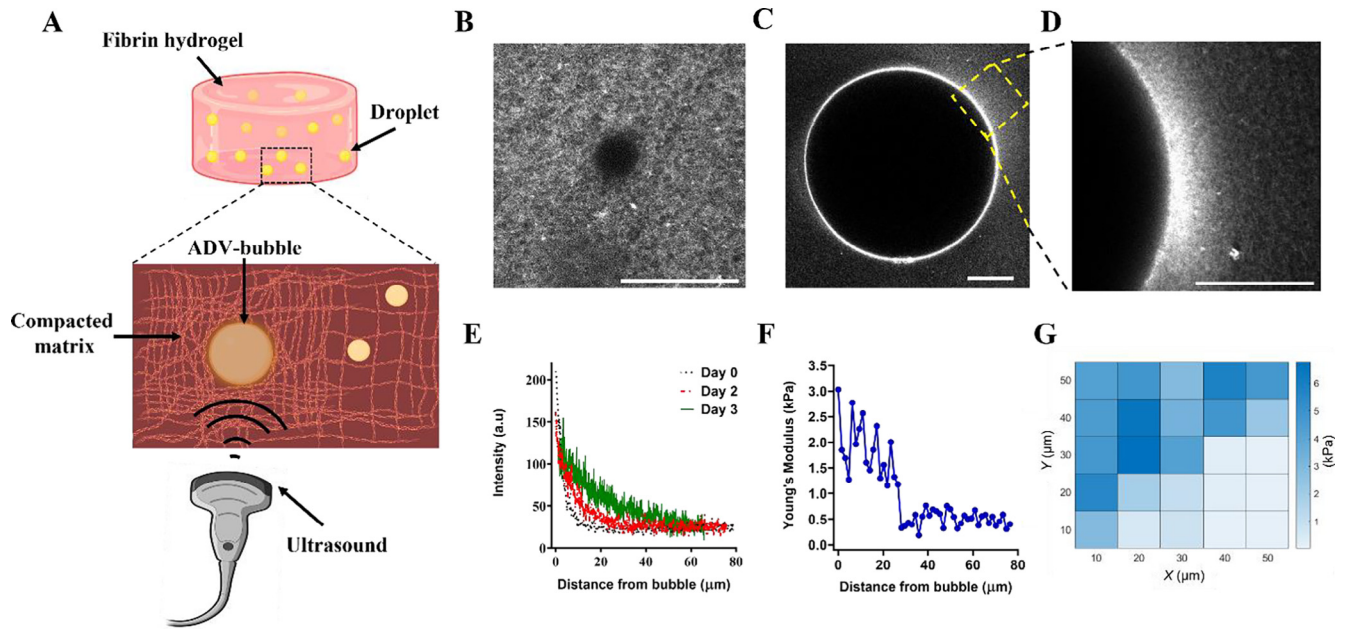


Figure 5. Acoustic droplet vaporization (ADV) in a hydrogel matrix can significantly affect its microstructural and micromechanical properties. ADV-induced changes have been studied in fibrin hydrogels using confocal microscopy and atomic force microscopy (A). Comparing confocal images of a phase-shift droplets before (B) and after (C) ADV indicates local compaction at the bubble-fibrin interface, as seen by an increase in fluorescence intensity that longitudinally persisted (E). Both one- (F) and two-dimensional (G) mapping of Young's moduli adjacent to the bubble were performed, showing ADV-induced stiffening proximal to the bubble-fibrin interface. Scale bar: 300 μm . Panels B, C, D, and G reprinted from Farrell, Aliabouzar [113], with permission from Elsevier. Panels E and F reprinted from Humphries, Aliabouzar [114] with permission from John Wiley and Sons.

onance frequency, which is dependent on bubble size [126,127]. Bubble-based approaches can provide material information at high resolution, both spatially and temporally, and at much higher strain rates (up to 10^6 s^{-1}) compared to conventional techniques ($<0.1 \text{ s}^{-1}$) [128]. The ability to characterize tissue response at high strain rates is also desirable for applications such as laser- or ultrasound-based surgical procedures as well as impact injuries [129,130].

Dependence of the ADV threshold on medium properties have also been documented. The ADV threshold of PFP droplets correlated with the surrounding fluid viscosity [106] and polyacrylamide gel stiffness [105]. The authors hypothesized that bulk fluids of higher viscosity may retard the expansion of the ADV nucleus, possibly driving recondensation of the vaporized nucleus at lower driving pressures. High-speed microscopy demonstrated that a 4-fold increase in fibrin density decreased the diameter of ADV-generated PFP bubbles by 2.5% and 35% during and after ADV, respectively [40] (Fig. 7). The above studies, though few, indicate that ADV has the potential to be utilized as an *in situ* microrheology probe. Since many tumors possess blood vessels permeable to submicron particles, it is possible that submicron phase-shift droplets could constitute a novel, super-resolution microrheology technique (Fig. 8). To better understand the

sensitivity of ADV to the surrounding medium rheology, the acoustic response of the generated bubbles, both the linear and nonlinear scattering components, should be systematically interrogated in soft gels of varying viscoelasticity. Furthermore, *in vitro* phantom studies as well as *in vivo* studies should be conducted to determine the size range as well as the distance the droplets can extravasate into the tissue.

5.2 ADV as an *in situ* pressure sensor

Monitoring local organ-level pressure can help determine the state of health as well as early diagnosis of disease related to the vascular system [131]. Clinically available techniques of *in situ* pressure measurement include catheter-based methods, which are invasive, or using Doppler ultrasound, which has issues with reproducibility. Using scattered emissions of contrast microbubbles for non-invasive pressure estimation was first proposed by Fairbank and Scully [132]. The authors showed theoretically and experimentally that the resonance frequency of air bubbles shifted with ambient pressure change in water. However, rapid dissolution of air bubbles (e.g., $\sim 20 \text{ ms}$ for a micron size air bubble) [74] and the necessity of monodisperse microbubbles for accurate pressure estimation prevented

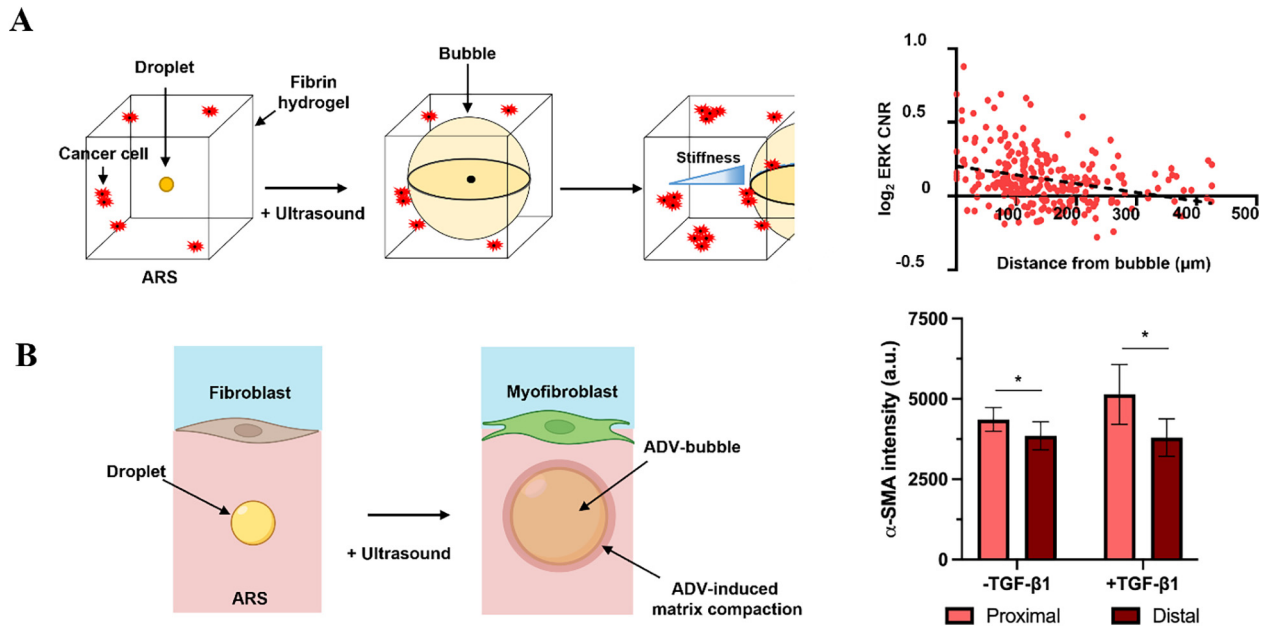


Figure 6. Matrix stiffening in fibrin-based acoustically responsive scaffolds (ARSs), induced by acoustic droplet vaporization (ADV), has been investigated in two separate biomedical applications. In one study, cell signaling was modulated in ARSs containing phase-shift droplets and triple negative breast cancer cells. Activities (i.e., cytoplasmic-to-nuclear ratio (CNR)) of extracellular signal-regulated kinase (ERK), one of the major kinases driving cancer progression, correlated inversely with distance from the ADV-induced bubble (A). In another study, the level of α -smooth muscle actin (α -SMA) expression was significantly elevated in human dermal fibroblasts proximal to bubbles compared to distal cells, irrespective of the addition of exogenous transforming growth factor- β 1 (TGF- β 1). Proximal and distal regions were defined as regions within 0.1 mm and greater than 0.5 mm from the bubble edge, respectively. Statistically significant differences ($p < 0.05$) are denoted by an asterisk (B). Panel A reprinted from Humphries, Aliabouzar [114], with permission John Wiley and Sons. Panel B reprinted from Farrell, Aliabouzar [113] with permission from Elsevier.

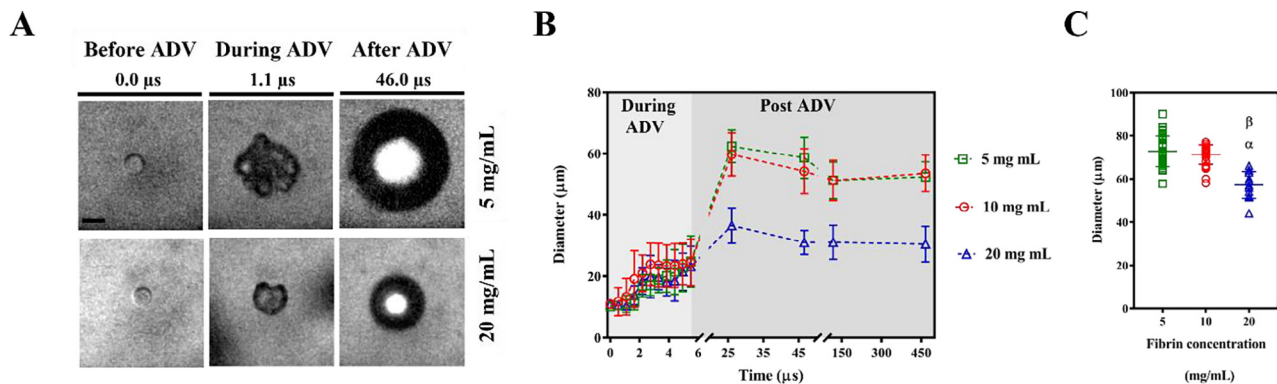


Figure 7. Ultra-high-speed images of a perfluoropentane (PFP) phase-shift droplet before, during, and after acoustic droplet vaporization (ADV) in acoustically responsive scaffolds with fibrin concentrations of 5 mg/mL and 20 mg/mL (A). The effect of fibrin concentration on the radial dynamics of the PFP droplet was quantified in microsecond timescales (B) as well as 30 s post-ADV (C) for varying fibrin concentrations. Statistically significant differences ($p < 0.05$) are denoted as α vs. 5 mg/mL fibrin and β vs. 10 mg/mL fibrin. Acoustic parameters were as follows: a single burst of 6 μs , 2.5 MHz, and 6.5 MPa peak rarefactional pressure. Exposure time of 100 ns was used. Scale bar: 15 μm . Panels A–C reprinted from Aliabouzar, Kripfgans [40], with permission from Elsevier.

clinical implementation. Later, Shi, Forsberg [133] experimentally found that the subharmonic component (i.e., response at frequency $f/2$ when excited at f) of the scat-

tered response from Levovist microbubbles, a first generation ultrasound contrast agent, was strongly dependent on ambient pressure. For a 25 kPa rise in the ambient pressure,

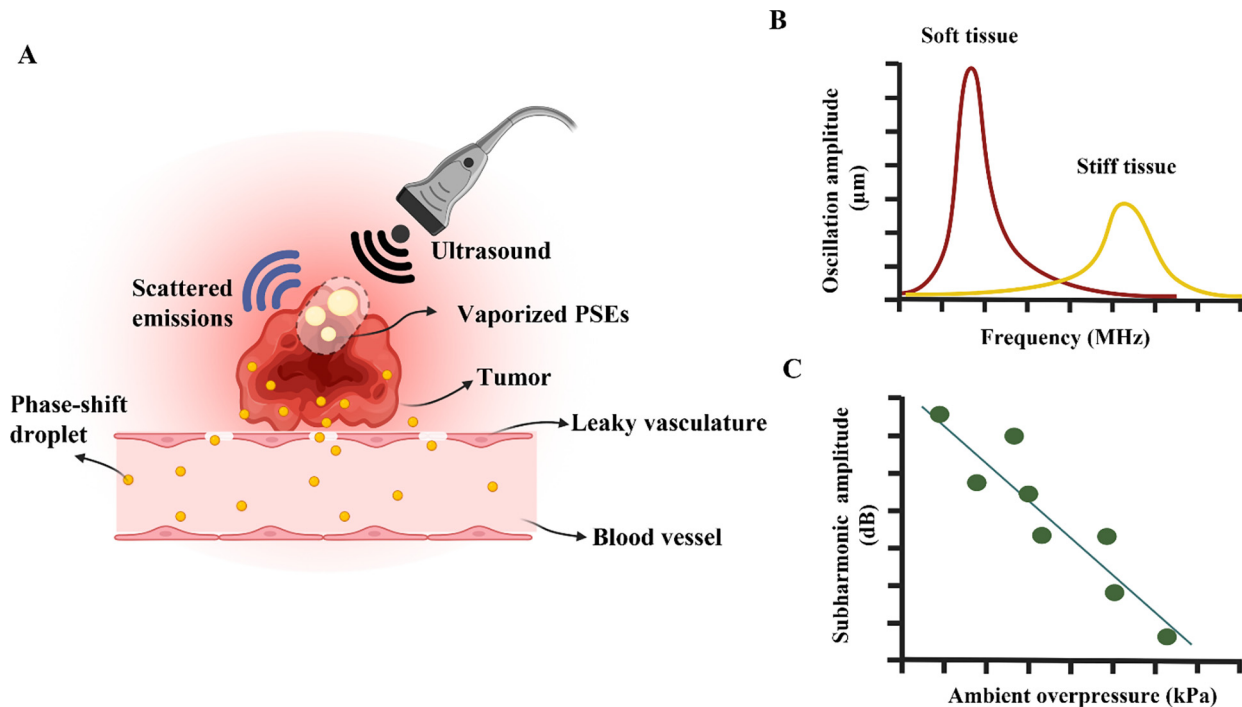


Figure 8. Potential application of acoustic droplet vaporization (ADV) as an *in situ* microrheometer or pressure sensor is schematically shown (A). Changes in the surrounding rheological properties may impact the oscillation amplitude or resonance frequency of ADV-generated bubbles (B). The reduction in the subharmonic response of ADV-generated bubbles as a function of ambient pressure may be used to monitor interstitial fluid pressure (C).

the authors reported a ~ 10 dB reduction in the amplitude of the subharmonic response.

Similar observations have been reported for other contrast microbubbles both *in vitro* and *in vivo* [134–136]. The simplified reasoning behind the reduction in the subharmonic response with increasing ambient pressure is the decrease in microbubble size and oscillation amplitude. However, Katiyar, Sarkar [137] numerically showed that subharmonic response depended on the frequency ratio – the ratio of the excitation frequency to the resonant frequency of the microbubbles. Therefore, increasing the ambient pressure changes the resonant frequency, and as a result the frequency ratio. Depending on the critical frequency ratio, an increasing or decreasing subharmonic response with the ambient overpressure can be expected. Non-invasive microbubble-based pressure estimation is limited to the vasculature due to the size of microbubbles. Therefore, it cannot be used for extravascular applications such as tumor interstitial pressure. In addition, microbubbles, with higher molecular weight gas cores, have a short circulation time (i.e., on the order of minutes) [138–140]. This highlights the need for a non-invasive microbubble-based approach for extravascular tumor pressure estimation.

Phase-shift droplets persist for much longer in the body (e.g., hours to days) and can also be prepared in submicron

sizes for extravascular interrogations [141]. *In vivo* studies have shown that submicron size PFP droplets have high stability in vascular circulation prior to ADV, as much as 4–5 h after intravenous injection in mice, and exhibited contrast enhancement of the tumor interstitial space post-ADV [142]. Unlike normal tissues, tumor tissues can have highly disorganized and leaky blood vessels that lead to poor lymphatic drainage and elevated interstitial pressure [143]. The reported size of inter-endothelial junctions in various tumors ranged between 380 nm and 2 μm [38,144], which can allow extravasation of submicron size phase-shift droplets. *In vivo* and clinical studies have shown that the interstitial pressure could reach up to 14 kPa higher in tumor tissues compared to healthy tissues [145]. This poses the question whether clinically relevant ambient overpressure is sufficient to impact the ADV threshold or the response of ADV-generated bubbles (Fig. 8).

Although hydrostatic overpressure may influence the superheat stability of phase-shift droplets, CNT predicts a negligible increase in the ADV threshold within physiologically relevant changes in the ambient pressure. According to classical nucleation theory, a change in the hydrostatic pressure from 0 to 17 kPa increased the ADV threshold only by 20 kPa [146]. However, experimentally measured ADV thresholds, based on echogenicity, showed a ~ 1.5 MPa

increase in the ADV threshold for a hydrostatic overpressure of ~ 12 kPa [147]. Since the ADV thresholds of phase-shift droplets, at the diagnostic ultrasound frequency range, are in the megapascal range, a physiologically relevant ambient overpressure should have negligible effect on the ADV threshold. However, the acoustic response of the ADV-generated bubbles can vary significantly under different ambient overpressure. Acoustic characteristics of the ADV-generated bubbles have only been investigated at ambient pressure. In a study by Aliabouzar, Kumar [148], the scattering spectrum of newly generated PFP and perfluorohexane bubbles depicted clear peaks at the fundamental frequency and the second harmonic indicating significant nonlinear components in the bubble oscillation. In the same study, subharmonic emission was only recorded for PFP droplets. The transient nature of vaporization (i.e., recondensation) of perfluorohexane droplets and therefore insufficient time for sustained oscillation was reported to be the reason for the lack of subharmonic response. Non-linearity of the acoustic response of ADV-generated PFP bubbles, was shown to be comparable to that of SonoVue contrast microbubbles in a study by Reznik, Lajoine [19]. Unlike compression-dominated oscillations of SonoVue microbubbles, generated PFP bubbles showed either expansion-dominated or symmetric oscillations depending on the size. Future studies, both experimental observations and theoretical modeling, are required to characterize the acoustic response of ADV bubbles under varying ambient overpressures and to provide insight into applicability of ADV as a non-invasive pressure estimation tool.

6 Summary

In this work we reviewed high-speed imaging studies and the predictive modeling that was undertaken over the past two decades to shed more light on ADV phenomena. Our understanding of the physics and underlying mechanisms of ADV has substantially improved, leading to the development of new diagnostic and therapeutic applications. We highlighted recent advances in tissue regeneration that have utilized ADV-induced stable or transient bubble formation as a tool for on-demand modulation of release parameters as well as tissue micromechanics. Based on experimental observations and theoretical modeling available in the literature, we discussed potential new directions of ADV in measuring rheological properties and pressure of the surrounding medium.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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